

68. HRPulse Workspace

Internal Human Resource Management Portal

Submitted By

Name: Samarth Navale

Roll Number: 150096725148

Branch: B.Tech Computer Science Engineering

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Institute: ITM

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Abstract

HRPulse Workspace is a React-based Human Resource Management portal developed to simplify employee administration tasks within an organization. The system provides employee directory management, organizational hierarchy visualization, leave request processing, salary rollback tracking, permission management, reporting chain tracking, tenure analysis, and automated HR task assignment.

The application uses CSV files as the primary data source and operates entirely on the client side without requiring a backend server or database. This approach keeps the system lightweight, portable, and easy to demonstrate while still covering all required HR workflows.

Problem Statement (68. HRpulse Workspace)

Develop an internal company website for HR staff to:

- Manage employee records
 - Track organizational hierarchy
 - Process leave requests
 - Maintain salary change history
 - Assign HR operational tasks
 - Manage role permissions
 - Analyze employee tenure
 - Track reporting relationships
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Objectives

The objectives of HRPulse Workspace are:

1. Create a centralized HR management dashboard.
2. Provide quick employee lookup and search capabilities.
3. Visualize company hierarchy through an interactive organizational chart.
4. Manage employee salary changes with rollback functionality.
5. Process leave requests using FIFO queue management.
6. Maintain role-based permission visibility.
7. Track reporting chains within the organization.
8. Automatically assign HR tasks to available staff.

Technology Stack

Frontend

- React.js
- JavaScript (ES6+)
- HTML5
- CSS3

Build Tool

- Vite

Data Processing

- PapaParse

UI Icons

- Lucide React

Data Source

- CSV Files

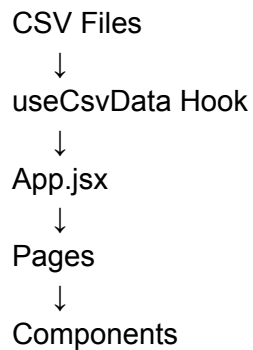
CSV Files Used

1. employees.csv
 2. leaveRequests.csv
 3. salaryHistory.csv
 4. roles.csv
 5. tasks.csv
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System Architecture

Architecture Overview

The application follows a centralized state architecture.

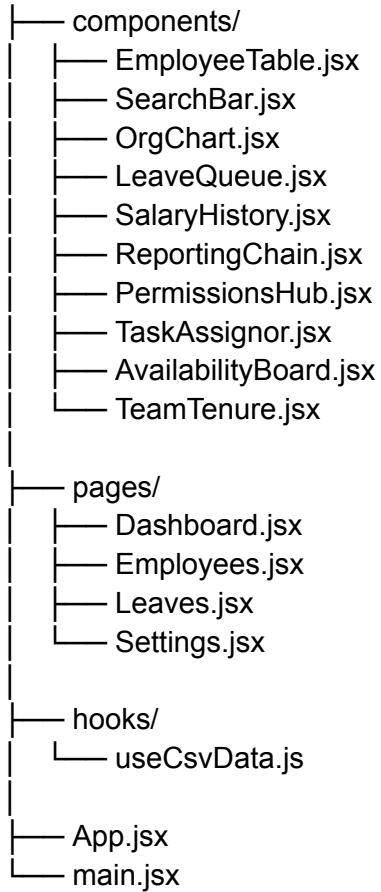


Data Flow

1. CSV files are loaded using PapaParse.
 2. Data is parsed into JavaScript objects.
 3. App.jsx stores all application state.
 4. Components receive data through props.
 5. User actions update state.
 6. Updated state automatically refreshes UI.
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Project Structure

src/



Features Implementation

A. Company Structure Viewer

Purpose

Display organizational hierarchy in a tree structure.

Implementation

- Recursive TreeNode component
 - Expand/Collapse support
 - Department visibility
 - Manager-Employee relationships
 - Circular reference protection
-

B. Salary Change Undo

Purpose

Track salary modifications and allow rollback.

Implementation

- Salary change logging
 - Previous salary preservation
 - Latest change undo functionality
 - Salary audit history
-

C. Time-Off Request Queue

Purpose

Process leave requests in exact submission order.

Implementation

- FIFO Queue
 - Pending Request List
 - Approve / Reject Actions
 - Processed Request History
 - Processing Timestamp Recording
-

D. Fast Employee Search

Purpose

Quickly locate employee information.

Implementation

- Search by Name
 - Search by Department
 - Search by Role
 - Debounced Search Input
 - Scrollable Employee Directory
-

E. Tenure Sorter

Purpose

Rank departments based on average employee tenure.

Implementation

- Department Grouping
 - Average Tenure Calculation
 - Descending Ranking
 - Team Comparison Table
-

F. Permissions Hub

Purpose

Display and manage role permissions.

Implementation

- Role-Permission Mapping
 - Permission Addition
 - Permission Removal
 - CSV-driven Permission Storage
-

G. Reporting Chain Tracker

Purpose

Identify the reporting path from employee to top management.

Implementation

- Manager Traversal Logic
- Circular Reference Protection
- Management Step Count
- Reporting Path Visualization

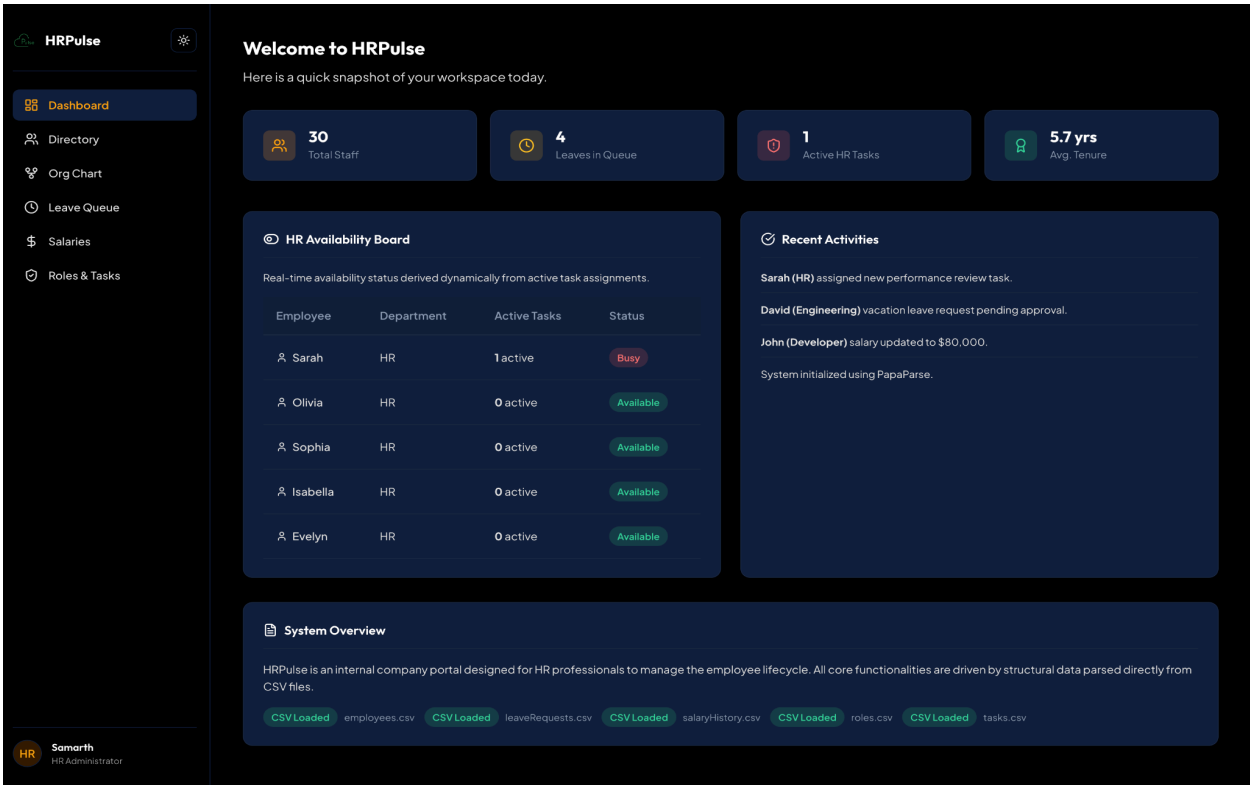
H. HR Task Assignor

Purpose

Automatically distribute HR responsibilities.

Implementation

- HR Availability Detection
- Round Robin Assignment
- Active Task Tracking
- Completed Task Tracking
- Dynamic Availability Calculation



 HRPulse

Dashboard

Directory

Org Chart

Leave Queue

Salaries

Roles & Tasks

HR Administrator

Employee Directory

Search, filter, update salaries, and track report chains.

Active Personnel

Search employee name, department, role...

All Departments

ID	Name	Department	Role	Salary	Join Date	Actions
#1	John	Engineering	Developer	\$80,000	2020-01-15	Update Salary
#2	Sarah	HR	HR Executive	\$60,000	2021-02-10	Update Salary
#3	David	Engineering	Developer	\$85,000	2019-08-12	Update Salary
#4	Alice	Management	CEO	\$150,000	2015-05-10	Update Salary
#5	Marcus	Engineering	Engineering Manager	\$110,000	2017-03-22	Update Salary
#6	Olivia	HR	HR Manager	\$95,000	2018-06-15	Update Salary
#7	Diana	Finance	Finance Manager	\$100,000	2016-11-01	Update Salary
#8	Emily	Engineering	QA Engineer	\$70,000	2021-09-01	Update Salary
#9	Michael	Engineering	Devops Specialist	\$90,000	2018-04-12	Update Salary

Reporting Chain Tracker

Team Tenure Sorter

 HRPulse

Dashboard

Directory

Org Chart

Leave Queue

Salaries

Roles & Tasks

HR Administrator

Reporting Chain Tracker

Find the reporting hierarchy and chain of command from any employee up to the CEO.

Select Employee

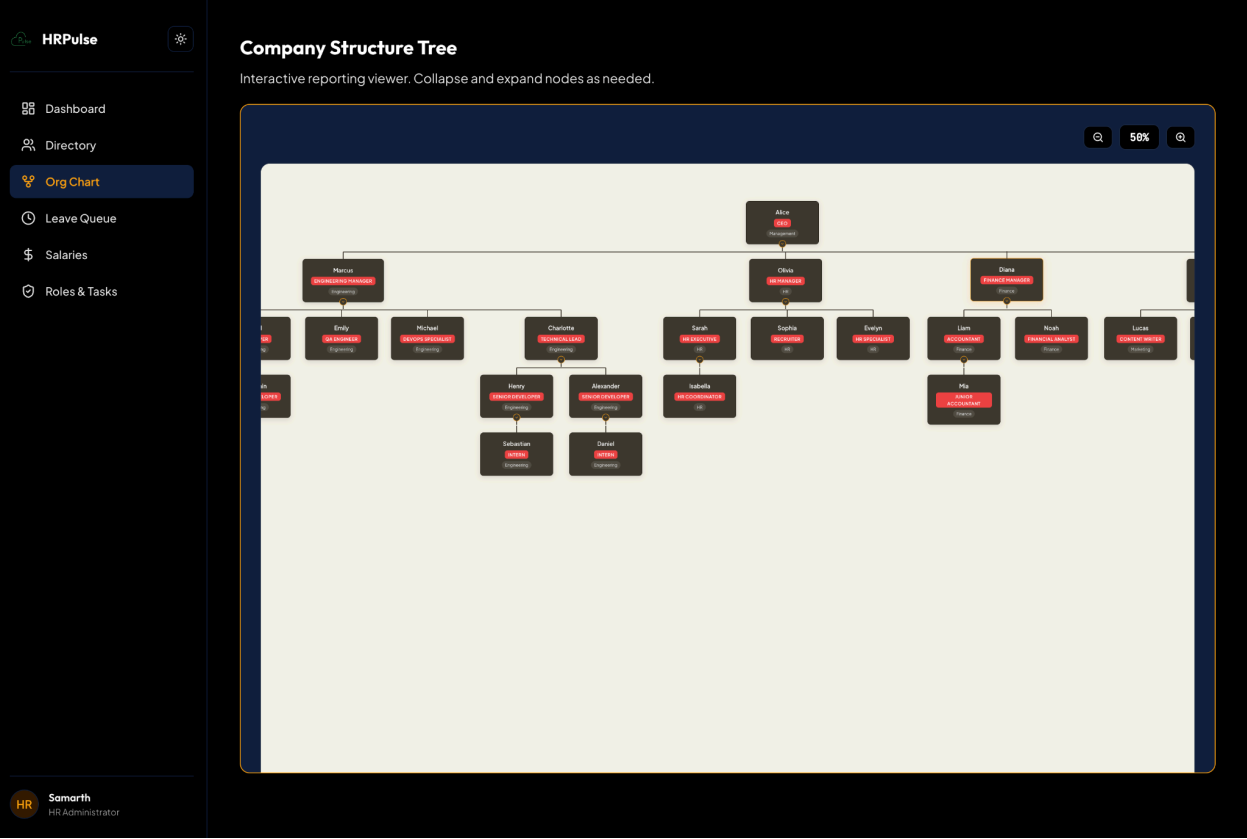
-- Choose Employee --

Select an employee above to calculate their reporting chain.

Team Tenure Sorter

Departments ranked by the average time staff members have been with the company.

Rank	Department	Average Tenure	Headcount	Trend
#1	Management	11.1 years	1 employee	High Retention
#2	Operations	6.3 years	3 employees	High Retention
#3	Finance	6.2 years	4 employees	High Retention
#4	HR	5.6 years	5 employees	High Retention
#5	Engineering	5.5 years	13 employees	High Retention
#6	Marketing	4.3 years	4 employees	High Retention



Dashboard

Directory

Org Chart

Leave Queue

Salaries

Roles & Tasks

HR

Samarth

HR Administrator

Submitted: 2025-01-05

Approve

Reject

William

#2

Employee ID: #16

Personal

Submitted: 2025-01-07

Waiting for front item...

Sarah

#3

Employee ID: #2

Maternity

Submitted: 2025-01-08

Waiting for front item...

Sebastian

#4

Employee ID: #24

Sick

Submitted: 2025-01-10

Waiting for front item...

Processed Requests Log

Employee	Type	Status	Processed Date
Noah	Vacation	Approved	-
John	Sick	Approved	-

Add to Queue (FIFO)

Dashboard

Directory

Org Chart

Leave Queue

Salaries

Roles & Tasks

HR

Samarth

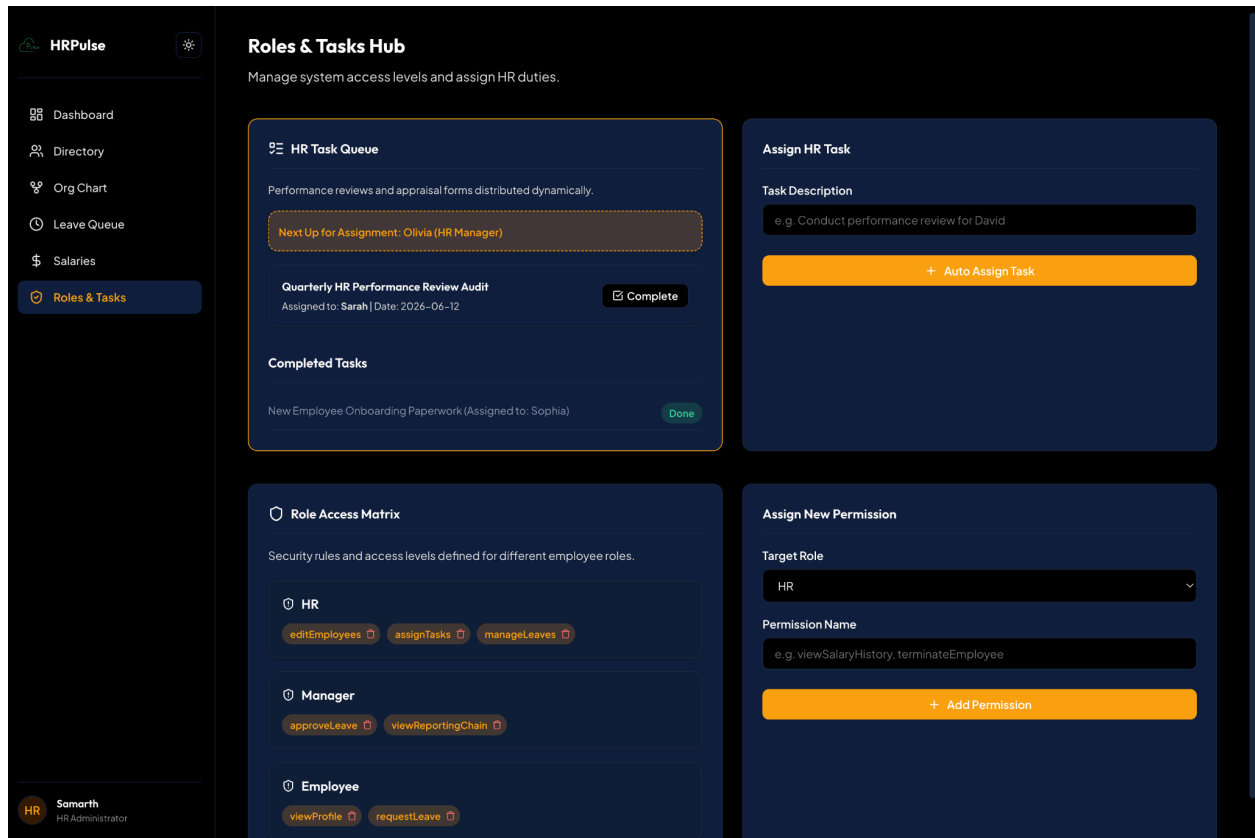
HR Administrator

Salary Change Log

Monitor, audit, and roll back recent salary adjustments in the workspace.

\$ Salary Change Log

Date	Employee	Old Salary	New Salary	Difference	Actions
2025-01-02	Charlotte	\$100,000	\$105,000	+\$5,000	Undo Change
2024-11-20	Michael	\$85,000	\$90,000	+\$5,000	Locked
2024-12-15	John	\$75,000	\$80,000	+\$5,000	Locked
2025-01-10	David	\$80,000	\$85,000	+\$5,000	Locked



Key Design Decisions

Why CSV Instead of Database?

- Lightweight setup
- Easy testing
- No server configuration
- Easy data inspection

Why React?

- Component-based architecture
- Reusable UI
- Efficient state updates

Why Derived Availability?

Availability is calculated from active tasks instead of being stored separately. This avoids synchronization issues and ensures consistent data.

Why No Redux?

React state hooks were sufficient.

Testing

The following modules were tested:

Module	Status
Employee Search	Passed
Salary Undo	Passed
Leave Queue FIFO	Passed
Org Chart	Passed
Reporting Chain	Passed
Permissions Hub	Passed
Task Assignment	Passed

Future Enhancements

1. Database Integration
 2. User Authentication
 3. Role-Based Access Control
 4. CSV Export Functionality
 5. Employee Profile Editing
 6. Real-Time Notifications
 7. Multi-Manager Reporting Structures
 8. Persistent Data Storage
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Conclusion

HRPulse Workspace successfully fulfills all specified assignment requirements. The application demonstrates practical usage of React components, state management, CSV-based data handling, recursive rendering, queue management, and derived-state computation.

The project provides a functional HR management portal while maintaining a simple and maintainable architecture suitable for academic evaluation and future expansion.

References

1. React Documentation
2. Vite Documentation
3. PapaParse Documentation

4. Lucide React Documentation

GitHub Repository

GitHub Link: <https://github.com/Samarth-ITM/HRpulse-Workspace>

Live Deployment

Live URL: <https://hrpulse-seven.vercel.app/>
